STUDY GUIDE & LEXICON

The Kinship Theater

Relatedness, inbreeding and identity by descent: the mechanisms, the estimators, and which methods work, why, and when. A companion to the narrated program.

19:22 running time **5** chapters **39** lexicon terms **45** references

I Identity by Descent 0:42

II The Reference Population 3:51

III Reading Relationships 6:37

IV Measuring Inbreeding 10:42

V Choosing Methods 15:44

Each chapter lists the key equations, the references and software it draws on, every numbered lexicon term from the video (with the time it appears), and self-check questions. Chapter V adds a decision table: which method fits which data and question.

I Identity by Descent

What is the one quantity beneath relatedness and inbreeding?

video 0:42–3:51

0:51 Alike in state, alike by descent · 1:33 The kinship coefficient · 2:15 Inbreeding is kinship, one generation back · 3:07 Relatedness, rescaled

KEY EQUATIONS

Kinship (coancestry)	φ_{ij} = P(one allele drawn from i and one from j, same locus, are IBD)
Inbreeding	$F_o = \varphi_{mf}$: an individual's inbreeding equals the kinship of its mother m and father f
Self-kinship	$\varphi_{ii} = (1 + F_i) / 2$
Path counting (Wright)	$\varphi_{ij} = \Sigma_{\text{paths}} (½)^n (1 + F_A)$, n = individuals in the path through common ancestor A, counting i and j
Standard kinships (outbred)	parent–offspring ¼ · full sibs ¼ · half sibs ⅛ · avuncular ⅛ · grandparent ⅛ · first cousins 1/16
Relatedness (Wright)	$r_{ij} = 2\varphi_{ij} / \sqrt{((1 + F_i)(1 + F_j))}$; $r = 2\varphi$ when outbred
Full sibs of related parents	parents with kinship f: $\varphi_{FS} = (1 + 2f)/4$, $r_{\text{FS}} = (1 + 2f) / (2(1 + f)) > ½$

KEY REFERENCES & TOOLS

SOURCE	TOOL OR TOPIC	WHAT IT SHOWS OR DOES
Wright 1922	Path coefficients	Defined the inbreeding coefficient and coefficient of relationship; path counting through common ancestors.
Malécot 1948	Probability of identity	Recast inbreeding and kinship as probabilities of identity by descent.

SOURCE	TOOL OR TOPIC	WHAT IT SHOWS OR DOES
Powell, Visscher & Goddard 2010	Review	Reconciles IBD and IBS: marker-based estimates are IBD relative to a reference population.
Speed & Balding 2015	Review	Relatedness in the genomic era; why allele-frequency weighting and the reference matter.

LEXICON

No. 001 0:52

Identity by state (IBS)

Two alleles with the same sequence, whatever their origin. This is what genotypes show directly.

No. 002 0:52

Identity by descent (IBD)

Two alleles that are copies of a single ancestral allele, inherited through the pedigree since a reference (base) generation.

No. 003 1:34

Kinship coefficient (φ)

Probability that alleles drawn at random, one from each of two individuals at the same locus, are IBD. Also called coancestry.

No. 004 2:02

Path counting

Wright's method: φ_ij = Σ over paths (½)^n (1 + F_A), with n individuals in each path through a common ancestor A.

No. 005 2:16

Inbreeding coefficient (F)

Probability that the two alleles at a locus within one individual are IBD. Equals the kinship of the individual's parents.

No. 006 2:42

Self-kinship

Kinship of an individual with itself: φ_ii = (1 + F_i) / 2.

No. 007 2:54

Kinship matrix

Matrix of all pairwise kinships; off-diagonal = pairwise coancestry, diagonal = (1 + F) / 2. A genomic relationship matrix is its marker-based estimate (scaled ×2).

No. 008 3:08

Relatedness (r)

Wright's coefficient of relationship: r = 2φ / √((1 + F_i)(1 + F_j)); equals 2φ for outbred individuals.

SELF-CHECK QUESTIONS

1. Compute the kinship of double first cousins by path counting. What is the inbreeding of their offspring?
2. Explain why the kinship matrix contains inbreeding on its diagonal. Why does this dissolve the apparent chicken-and-egg problem?
3. Why can a correctly working estimator report full-sib relatedness above 0.5?

II The Reference Population

Identity by descent, relative to what?

video 3:51–6:37

3:58 Identical by descent, since when? · 4:46 Rare alleles carry the signal · 5:34 When structure masquerades as kinship

KEY EQUATIONS

Genomic relationship matrix	$A_{ij} = (1/L)\,\Sigma_l\,(x_{il} - 2p_l)(x_{jl} - 2p_l)\,/\,[2p_l(1-p_l)]$; $x =$ allele dosage 0, 1, 2
What the GRM estimates	off-diagonal $\approx 2\varphi_{ij} = r_{ij}$ (outbred); diagonal $\approx 1 + F_i$
Relative estimates	mean pairwise estimate in the reference sample ≈ 0, so pairs less related than average are negative
KING-robust (between-family)	$\varphi_{ij} = (N_{Aa,Aa} - 2N_{AA,aa})\,/\,(2N_{Aa}{}^{(i)}) + ½ - ¼\,(N_{Aa}{}^{(i)} + N_{Aa}{}^{(j)})\,/\,N_{Aa}{}^{(i)}$, $N_{Aa}{}^{(i)} \le N_{Aa}{}^{(j)}$: shared heterozygotes and opposite homozygotes, no allele frequencies
Wahlund effect	pooling groups with allele frequencies p_k: heterozygote deficit $= 2\,\mathrm{Var}(p_k)$, so apparent $F_{IS} = F_{ST}$

KEY REFERENCES & TOOLS

SOURCE	TOOL OR TOPIC	WHAT IT SHOWS OR DOES
Queller & Goodnight 1989	Moment estimator	Allele-frequency-weighted relatedness from codominant markers.
Lynch & Ritland 1999; Wang 2002	Moment estimators	Alternative weightings; differ in bias and variance by locus number and allele frequency.
Wang 2014	Assessment	Compares relatedness and inbreeding estimators, including behaviour with inbred individuals.

SOURCE	TOOL OR TOPIC	WHAT IT SHOWS OR DOES
Yang et al. 2010, 2011	GRM · GCTA	Genomic relationship matrix and individual F estimators (F_hom, F_GRM, F_UNI).
Manichaikul et al. 2010	KING	Relationship inference robust to population structure.
Conomos et al. 2015, 2016	PC-AiR · PC-Relate	Ancestry from an unrelated subset; kinship and F with individual-specific allele frequencies.
Alexander et al. 2009; Frichot et al. 2014	ADMIXTURE · sNMF	Estimating clusters and ancestry proportions before kinship analysis.

LEXICON

No. 009 3:59

Base population

The reference generation whose members are defined as unrelated and non-inbred; IBD is counted only since then.

No. 010 4:25

Negative relatedness

A marker estimate below zero: the pair shares fewer alleles than the average pair in the reference sample. Expected, not an error.

No. 011 4:58

Moment estimator

Kinship estimator built from allele-frequency-weighted IBS sharing (Queller–Goodnight, Lynch–Ritland, Wang, GRM). Fast; relative to the reference frequencies.

No. 012 4:58

Genomic relationship matrix (GRM)

$A_{ij} = (1/L) \Sigma_l (x_{il} - 2p_l)(x_{jl} - 2p_l) / 2p_l(1 - p_l)$; x = allele dosage (0, 1, 2). Estimates 2φ off the diagonal, 1 + F on it.

No. 013 5:56

Wahlund effect

Heterozygote deficit created by pooling genetically differentiated groups; mimics inbreeding and inflates apparent kinship within groups.

No. 014 6:06

KING-robust

Kinship estimator (Manichaikul et al. 2010) using only the pair's own genotypes (heterozygote sharing), robust to structure among pairs; assumes each pair is outbred and shares one ancestry.

No. 015 6:20

PC-AiR / PC-Relate

Conomos et al. (2015, 2016): PCs from an unrelated subset define ancestry; kinship and F are then estimated with individual-specific allele frequencies.

SELF-CHECK QUESTIONS

1. Why do marker-based kinship estimates take negative values? What is the base population of a GRM?

2. Sketch how pooling two differentiated populations inflates both kinship within populations and apparent inbreeding.
3. Describe the PC-AiR / PC-Relate loop and explain why it turns the chicken-and-egg problem into an iteration.

III Reading Relationships

What can a pair's genotypes tell you, and what can they not?

video 6:37–10:42

6:45 Cotterman's triangle · 7:43 Nine ways to share · 8:46 Segments, not loci · 9:44 Asking a relative question

KEY EQUATIONS

Cotterman coefficients	$k_0 + k_1 + k_2 = 1;\ \ \varphi = k_1/4 + k_2/2$
Standard (k0, k1, k2)	unrelated (1, 0, 0) · parent–offspring (0, 1, 0) · full sibs (¼, ½, ¼) · half sibs, avuncular, grandparent (½, ½, 0) · first cousins (¾, ¼, 0)
Jacquard kinship	$\varphi = \Delta_1 + ½(\Delta_3 + \Delta_5 + \Delta_7) + ¼\Delta_8;\ \ \text{outbred: } \Delta_7, \Delta_8, \Delta_9 = k_2, k_1, k_0$
Inbreeding from Jacquard	$F_i = \Delta_1 + \Delta_2 + \Delta_3 + \Delta_4;\ \ F_j = \Delta_1 + \Delta_2 + \Delta_5 + \Delta_6$
IBD segment length	segments separated by m meioses have mean length $\approx 100/m$ cM; grandparent (1 meiosis) > half sibs (2) > avuncular (3)
Realized full-sib sharing	expected ½; human SD ≈ 0.04 (≈ 0.4–0.6); wider with shorter genetic maps

KEY REFERENCES & TOOLS

SOURCE	TOOL OR TOPIC	WHAT IT SHOWS OR DOES
Cotterman 1940; Thompson 1975	k-space	Three IBD-sharing coefficients and likelihood estimation of pairwise relationship.
Jacquard 1974	Condensed identity states	Nine states for a pair when inbreeding is allowed.
Hill & Weir 2011; Thompson 2013	Realized IBD	Variance of actual sharing from Mendelian sampling and linkage.
Visscher et al. 2006	Human full sibs	Measured realized genome sharing between full siblings.

SOURCE	TOOL OR TOPIC	WHAT IT SHOWS OR DOES
Hanghøj et al. 2019	NgsRelate v2	ML estimates of nine Jacquard coefficients and F from genotype likelihoods.
Pew et al. 2015	related (R)	Seven estimators; coancestry(allow.inbreeding = TRUE); familysim for simulation.
Smouse & Peakall 1999; Banks & Peakall 2012	Spatial autocorrelation	Distance-class kinship by pair class detects sex-biased dispersal.
Goudet, Perrin & Waser 2002	Sex-biased dispersal	Tests comparing the sexes using biparentally inherited markers.

LEXICON

No. 016 6:46

Cotterman coefficients (k0, kl, k2)

For outbred pairs, the probabilities of sharing 0, 1 or 2 alleles IBD at a locus. φ = k1/4 + k2/2.

No. 017 7:20

Second-degree relatives

Half sibs, avuncular and grandparent–grandchild pairs: identical (k0, k1, k2) = (½, ½, 0), so indistinguishable with unlinked markers.

No. 018 7:52

Jacquard coefficients (Δ1–Δ9)

Probabilities of the nine condensed IBD states among the four alleles of a pair, allowing inbreeding. Δ7, Δ8, Δ9 = k2, k1, k0 when outbred.

No. 019 8:36

NgsRelate v2

Hanghøj et al. (2019): ML estimates of all nine Jacquard coefficients and F from genotype likelihoods; suited to low-depth data.

No. 020 8:47

IBD segment

A chromosome stretch inherited intact from a common ancestor. Expected length shrinks with the number of meioses separating the pair.

No. 021 9:14

Realized relatedness

The fraction of the genome actually shared IBD, which varies around the pedigree expectation because of Mendelian sampling.

No. 022 9:45

Sex-biased dispersal

One sex disperses farther than the other; predicts higher kinship among nearby members of the philopatric sex.

No. 023 10:17

Permutation test

Builds a null distribution by shuffling labels (here, sex) and recomputing the statistic; robust to shared baseline biases.

SELF-CHECK QUESTIONS

1. Place parent–offspring, full sibs, half sibs, and unrelated pairs in Cotterman space. Which pairs cannot be separated with unlinked markers, and what can separate them?

2. Why does forcing an inbred pair into a three-coefficient model risk misclassification and biased kinship?

3. In a test for sex-biased dispersal, when does a shared baseline bias cancel, and when might it not?

IV Measuring Inbreeding

What does heterozygosity hide, and what do runs of homozygosity reveal?

video 10:42–15:44

10:50 Inbred, yet in equilibrium · 11:48 What fakes inbreeding · 12:38 Does inbreeding vary? · 13:33 Runs of homozygosity · 14:30 Reading runs responsibly

KEY EQUATIONS

F_IS	$F_{IS} = 1 - H_O / H_E$: non-random mating relative to the current population
Drift inbreeding	$F_t = 1 - (1 - 1/2N_e)^t$; $H_t = H_0(1 - 1/2N_e)^t$
Missing an allele	P(only one allele seen at depth d | heterozygote) $= 2(½)^d$
g2 (identity disequilibrium)	standardized covariance in heterozygosity between pairs of loci (David et al. 2007); g2 > 0 ⇒ variance in F
F_ROH	$F_{ROH} = \Sigma$ length of runs ≥ threshold / length of autosomal genome
Run length and age	common ancestor g generations back: mean run length ≈ 100 / $(2g)$ cM; half-sib mating ⇒ $F = ⅛$

KEY REFERENCES & TOOLS

SOURCE	TOOL OR TOPIC	WHAT IT SHOWS OR DOES
Wahlund 1928	Population mixture	Pooling subdivided populations creates a heterozygote deficit.
David et al. 2007; Szulkin et al. 2010	g2 · HFCs	Identity disequilibrium and why heterozygosity–fitness correlations need variance in F.

SOURCE	TOOL OR TOPIC	WHAT IT SHOWS OR DOES
Stoffel et al. 2016	inbreedR	g2 with bootstrap CIs; expected HFC strength.
Kardos et al. 2015, 2016	Genomic inbreeding	ROH-based F outperforms pedigree F in the wild.
Huisman et al. 2016	Red deer, Rum	Inbreeding depression across the lifespan measured with genomic F.
Ceballos et al. 2018	ROH review	Run length classes as windows into population history.
Meyermans et al. 2020	PLINK ROH	Parameter choices for medium-density SNP data.
Narasimhan et al. 2016; Druet & Gautier 2017; Bertrand et al. 2019	bcftools roh · RZooRoH	HMM ROH calling; autozygosity partitioned by age class.
Vieira et al. 2016; Renaud et al. 2019	ngsF-HMM · ROHan	IBD tracts and heterozygosity from low-coverage data.

LEXICON

No. 024 10:51

F_IS

1 − H_O / H_E: heterozygote deficit relative to Hardy–Weinberg expectation in the current sample; measures non-random mating, not total inbreeding.

No. 025 11:12

Drift inbreeding

Inbreeding that accumulates in a finite population under random mating: F_t = 1 − (1 − 1/2N_e)^t. Invisible to F_IS.

No. 026 11:58

Allelic dropout

Failure to observe one allele of a heterozygote (low depth, null alleles), producing false homozygotes and spurious inbreeding.

No. 027 12:10

Genotype likelihood

Probability of the sequencing reads under each possible genotype; carries depth-dependent uncertainty forward instead of forcing a hard call.

No. 028 12:39

Marker-based F

Individual F from genotypes: F_hom (PLINK --het), F_GRM and F_UNI (GCTA). Relative to sample allele frequencies; noisy with few loci.

No. 029 13:06

Identity disequilibrium (g2)

Correlation of heterozygosity across loci caused by variance in inbreeding among individuals; g2 > 0 means F varies (Stoffel et al. 2016, inbreedR).

No. 030 13:34

Autozygosity

Homozygosity because both alleles are IBD copies of one ancestral allele (as opposed to alike in state by chance).

No. 031 13:46

F_ROH

Proportion of the (autosomal) genome in runs of homozygosity above a length threshold; a realized, largely frequency-free measure of inbreeding.

No. 032 14:07

ROH length and age

A common ancestor g generations back gives runs of expected length ≈ 100 / 2g cM: long runs = recent inbreeding, short runs = ancient.

No. 033 14:41

ROH length classes

Partitioning F_ROH by run length (e.g. < 2, 2–8, > 8 cM) to separate ancient background autozygosity from recent inbreeding.

No. 034 15:07

ROH callers

bcftools roh (HMM, genotype likelihoods), RZooRoH (multi-class HMM by age), ngsF-HMM and ROHan (low coverage), PLINK --homozyg (sliding window).

SELF-CHECK QUESTIONS

1. Explain how a bottlenecked population can be highly inbred with F_IS ≈ 0.
2. List three ways a heterozygote deficit can be manufactured, and a diagnostic for each.
3. What does g2 test, and why should you check it before reporting individual F?
4. Why does F_ROH largely sidestep the base-population problem? What data does it require?

V Choosing Methods

Which tool, for which data and which question?

video 15:44–19:22

15:51 Let the pedigree generate the sharing · 16:40 Which method, when · 17:35 Simulate your own power · 18:31 The egg was never the problem

WHICH METHOD, WHEN

QUESTION	AMPLICON PANEL (10^2–10^3 SNPS)	RADSEQ	WHOLE GENOMES
Mean kinship among classes (e.g., sex-biased dispersal)	GRM or moment estimators + permutation	Same	Same
Parentage & sibships	COLONY, sequoia (+ ages)	COLONY, sequoia	Likelihood pedigree methods
Separate half sibs, avuncular, grandparent	Only with ages / pedigree context	Only with ages / pedigree context	IBD segments on a chromosome-level assembly
Individual inbreeding	Noisy marker F; check g2 first	Likelihood heterozygosity; cautious ROH	F_ROH by length class
Always first	Structure: analyze within clusters or use KING-robust / PC-Relate		

KEY REFERENCES & TOOLS

SOURCE	TOOL OR TOPIC	WHAT IT SHOWS OR DOES
Jones & Wang 2010	COLONY	Maximum-likelihood sibship and parentage reconstruction; optional inbreeding.
Huisman 2017	sequoia	Pedigree reconstruction from SNP panels with birth years and age priors; EstConf for error rates.
Baetscher et al. 2018	CKMRsim	False-positive and false-negative rates for kin categories by simulation.
Pew et al. 2015	related · familysim	Simulate known relatives with your allele frequencies to compare estimators.

LEXICON

No. 035 15:52

Pedigree likelihood

P(all genotypes | pedigree): candidate pedigrees are scored jointly, so inbreeding loops are handled by construction.

No. 036 16:12

COLONY

Jones & Wang (2010): ML sibship and parentage reconstruction; tolerates genotyping error; optional inbreeding.

No. 037 16:12

sequoia

Huisman (2017): pedigree reconstruction from SNP panels (hundreds of loci), using birth years and age priors.

No. 038 17:36

Simulation validation

Simulating known pedigrees with your own allele frequencies and markers to measure the bias, precision and error rates of your estimators.

No. 039 18:00

Misclassification rate

Fraction of pairs of one true relationship assigned to another; the overlap of simulated estimate distributions.

SELF-CHECK QUESTIONS

1. For your own study design, which cells of the decision table apply, and what is the limiting factor?
2. Design a simulation to test whether your marker panel can distinguish full from half siblings at plausible inbreeding.
3. Write the six-step workflow from memory and justify the order.

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